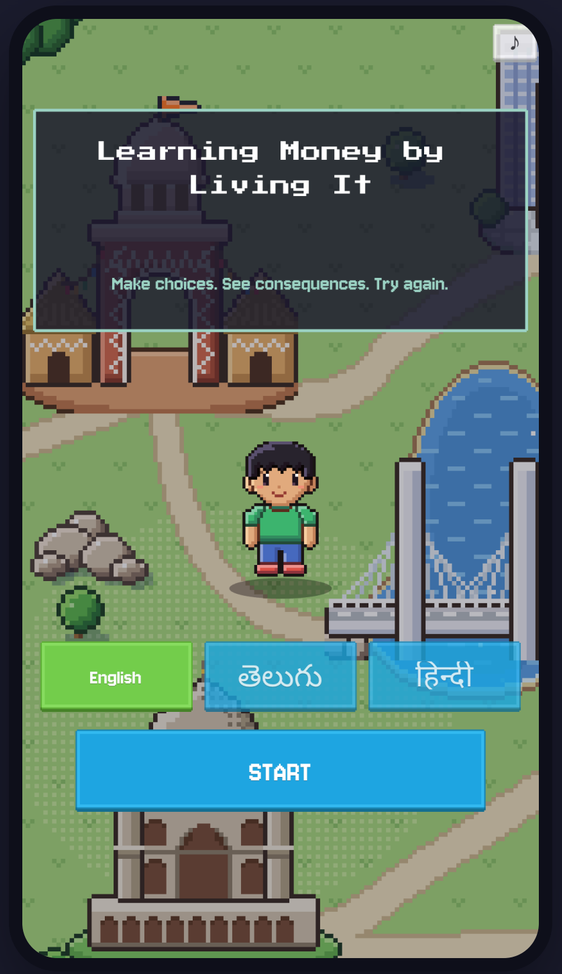
MEGATHON X · VANGUARD PROBLEM STATEMENT

# Learning Money by Living It

A pixel-art life simulation that teaches financial literacy to Intermediate students in Hyderabad by letting them live it.

**Team 97**

English · తెలుగు · हिन्दी



THE PROBLEM

# What the brief asks for

**A digital experience that Grade 11-12 students from underserved communities across Hyderabad actually want, with the theme of financial literacy.**

Start with about 1,500 students through NGOs, and design to scale. Keep it educational, scalable and privacy-conscious.

Learn by doing: **make choices, see consequences, try again.**

01
### Plan and budget
Needs vs wants, choices, goals

02
### Save and manage
Habits, spending, unexpected needs

03
### Understand investing
Risk, time, compounding, diversification

04
### Stay financially safe
Protect information, spot suspicious offers

**Guardrail: educate, do not recommend investments. Simulations are assumptions, not promises.**

OUR READING OF THE PROBLEM

# Financial literacy here means their own money

### Their money

Pocket money from parents, a TS ePASS scholarship landing in a first bank account, wages from a shop job, UPI on a family phone, a first salary a few years away.

### Their traps

Impulse buys on food apps, "no-cost" EMIs, lending to friends, fake scholarship officers, OTP requests, instant-loan apps, fake payment screenshots.

### Their setting

NGO sessions, shared low-end Android phones, patchy internet, and homes that speak Telugu, Hindi or English.

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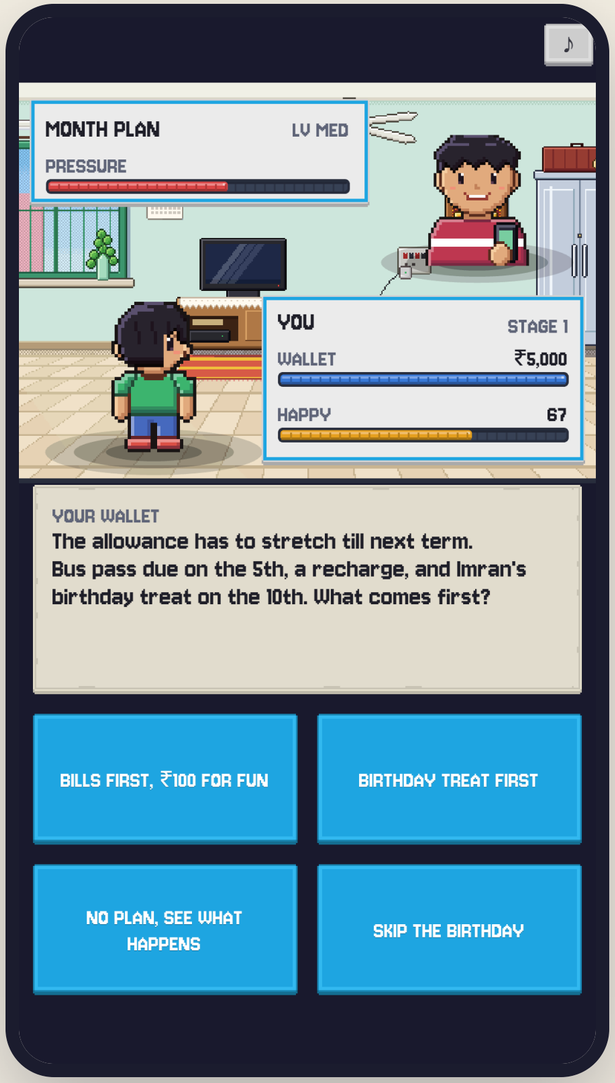
**Two easy answers miss:** a quiz app with coins is a textbook with confetti, and a stock-trading simulator breaks the guardrail and looks nothing like a 16-year-old's month.

WHAT WE BUILT

# Live a life, one money battle at a time

A browser game that works offline on cheap phones. The student lives from 16 to 60 across three life stages, and **every money decision is a Pokémon-style battle** at a Hyderabad place they know.

1. The opponent is the situation: a scam call, a festival sale, a "no-cost" EMI
2. Wallet and happiness bars move the moment you choose
3. Then the Professor walks in: praise, or a lesson and a short animation
4. Mistakes come back later, in disguise, until you get them right



ONE PLAYTHROUGH

# From handover to report, in one sitting

1
**Setup**
Student, or an NGO facilitator with a batch code and consent

2
**You**
Avatar, nickname, and 4 picture questions that pick a persona

3
**Quiz**
5 questions: the before score and the difficulty

4
**Student**
Stage 1, age 16: pocket money, 4 months

5
**First salary**
Stage 2, about 22: rent, first investing, 3 months

6
**Earning more**
Stage 3, about 30: EMIs, loans, bigger scams, 3 months

7
**Age 60**
Epilogue, what if, after-quiz, PDF report

**Personas change the story, never the difficulty.** Pocket money, scholarship or earner: the same scam is an OTP text on mom's phone, a fake scholarship officer, or a fake payment screenshot at the shop.

**Stage jumps skip 5, 6 and 25 years** so compounding is visible, and money only comes from life: allowance, scholarship, wages.

WHY THIS IS THE RIGHT DESIGN

# Six decisions, each for a reason

### Consequence first, lesson second

The bars move before anyone explains, so the lesson lands on a choice the student already owns.

### Mistakes return in disguise

A weak choice goes into a mistake bank and comes back in a new situation until it is answered right.

### Real life is delayed

45 delayed effects: the EMI bites every month, the loan to a friend never comes back.

### Prices match their pocket money

A ₹200 bat costs the same share of in-game money as of their real pocket money.

### Same lesson, their story

Three personas reframe every task. Only the quiz decides how hard the game is.

### Measured, not assumed

A before and after quiz in new situations, an NCFE-aligned score and a safety score.

ALL FOUR PILLARS

# 17 concepts, each taught by a task

### PLAN AND BUDGET · 4

- Needs vs wants
- Plan the month
- Compare before buying
- Borrowing costs extra

*"Biryani for ₹300 in 20 minutes, or chips for ₹20?"*

### SAVE AND MANAGE · 3

- Save first, spend after
- Emergency buffer
- Named savings goal

*"Lend me ₹500? I'll return it Monday, promise."*

### INVESTING · 5

- Money now vs later
- Compounding
- Risk and return
- Diversification
- Inflation

*"Banyan Savings, Chai Stall Fund or Mango Orchard Co.?"*

### STAY SAFE · 5

- Spot fake officials
- Never share OTP or PIN
- Too good to be true
- Instant-loan traps
- Fake payment proof

*"Pay ₹500 now or your scholarship is cancelled!"*

24 main tasks across the three stages, 72 persona framings, 22 detours and 16 random events. Every concept has an explainer animation and a replay in the library.

EVERY SUCCESS MEASURE

# Each requirement, and where it lives

| Requirement | What the game does |
|---|---|
| **Simple** | One decision per screen, 2 to 4 big buttons, a one-line lesson after each choice |
| **Relatable** | 13 Hyderabad places on 20 original backdrops; prices scaled to each student's pocket money |
| **Persona-led** | 3 personas reframe all 24 tasks (72 framings); 3 difficulty levels; life stages from 16 to 60 |
| **Language-ready** | English, Telugu and Hindi; a spoken Professor; a new language is one new string file |
| **Game-based** | Before and after quiz, NCFE-aligned score and safety score, all measured inside the game |
| **Intuitive UI** | Tap a landmark to travel; Info panel; replay library; hints after two fails; sound and voice switches |
| **NGO-integrated** | Facilitator setup with batch code and consent; PDF report to the facilitator, guardian in CC |
| **Interactive, fun** | Money battles, a living city map, scam calls that ring, mini-games, a turf match you earn |
| **Safe to explore** | In-game rupees only; detours and restarts instead of game over |
| **Guardrail** | Made-up assets and companies only; nothing recommended; every projection labelled an assumption |
| **Privacy** | Nickname only; never asks for OTP, PIN, Aadhaar, or a student's phone or email |

INTERACTIVE AND FUN

# Side gigs that teach, and a match you earn

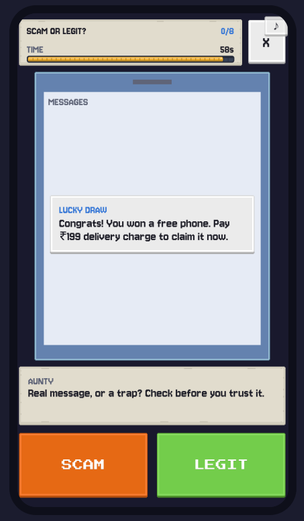
Mini-games pay 8% of a month's income and +10 happiness, and lock while your bars are low. Saving pays off too.




## Kirana Rush

**Needs vs wants**

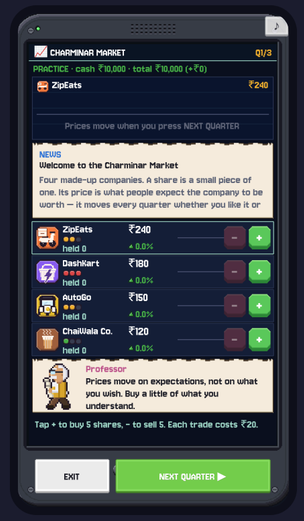
Rush hour at the kirana: sort each item as NEED or WANT.




## Scam or Legit?

**Spot scams**

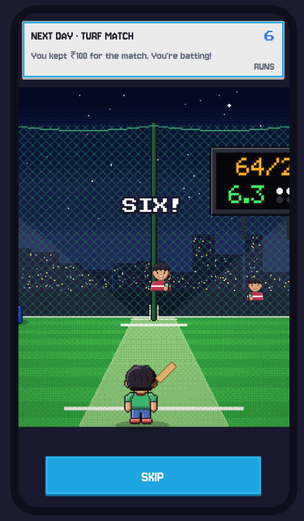
Help an aunty sort her phone messages. Swipe the fakes.




## Charminar Market

**Risk, diversification**

Made-up companies, a monsoon, a hot tip and a crash.




## The turf match

**Earned by saving**

Pick ₹20 chips over ₹300 biryani and you bat tomorrow.

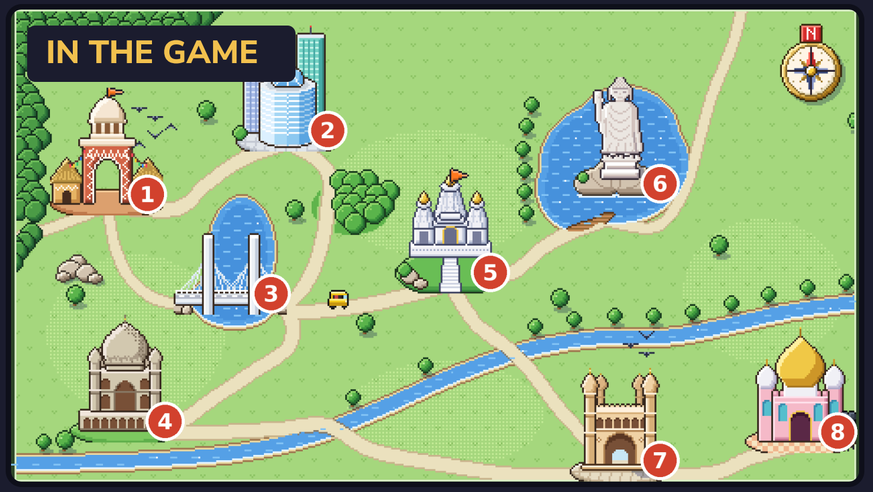
PART TWO

# Under the hood

A map that really is Hyderabad, analogies from home, an economy sized to each student, consequences that follow you, and a market that scores behaviour, not luck.

COOL FACT

# Our map is Hyderabad







1 Shilparamam · 2 HITEC City · 3 Durgam Cheruvu · 4 Qutb Shahi Tombs · 5 Birla Mandir · 6 Hussain Sagar · 7 Charminar · 8 Ramoji Film City

## 47 of 55

north-south and east-west comparisons between landmark pairs match the real city

## 78.47° E

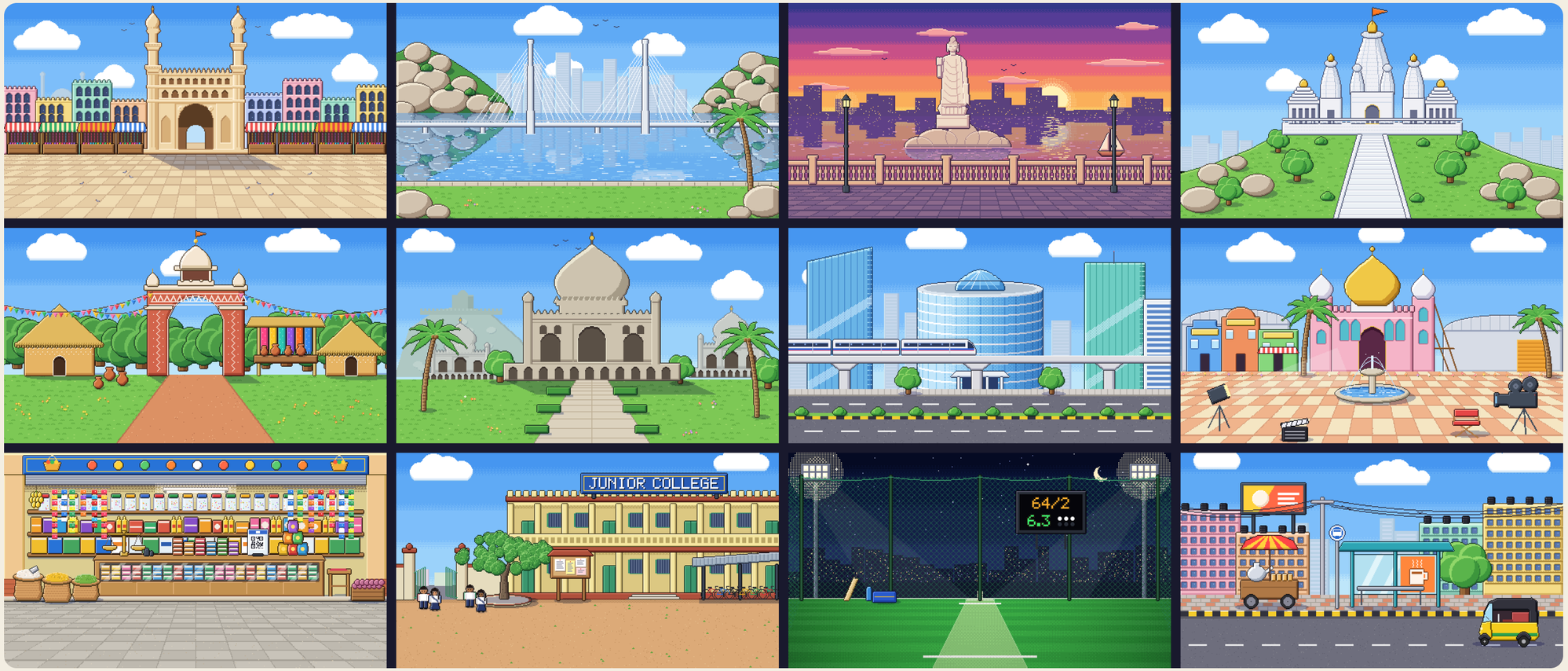
Charminar (7) is due south of Hussain Sagar (6): one column in the game, with the Musi between

## West to east

the IT corridor (1 to 3) in the west, the old city in the south, Ramoji (8) far out east

20 ORIGINAL HYDERABAD BACKDROPS

# Every battle happens somewhere they know



IT FEELS LIKE A REAL PHONE GAME

# Small touches that make it feel real

### Scam calls ring

A dodgy caller rings like a real phone. Decline, and he rings straight back, because scammers don't give up.

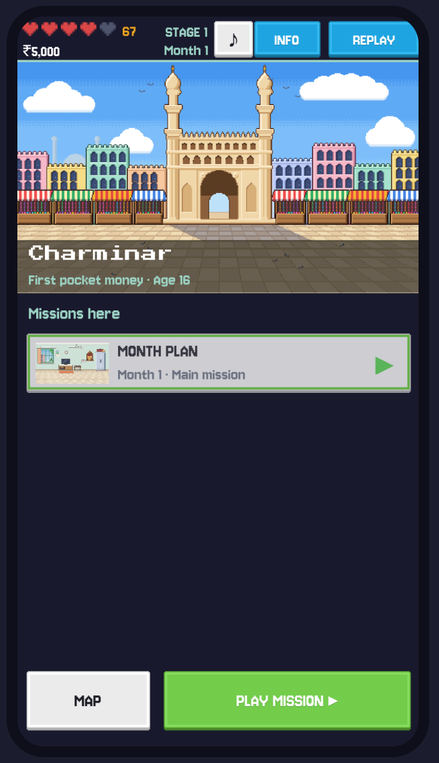
### A living city

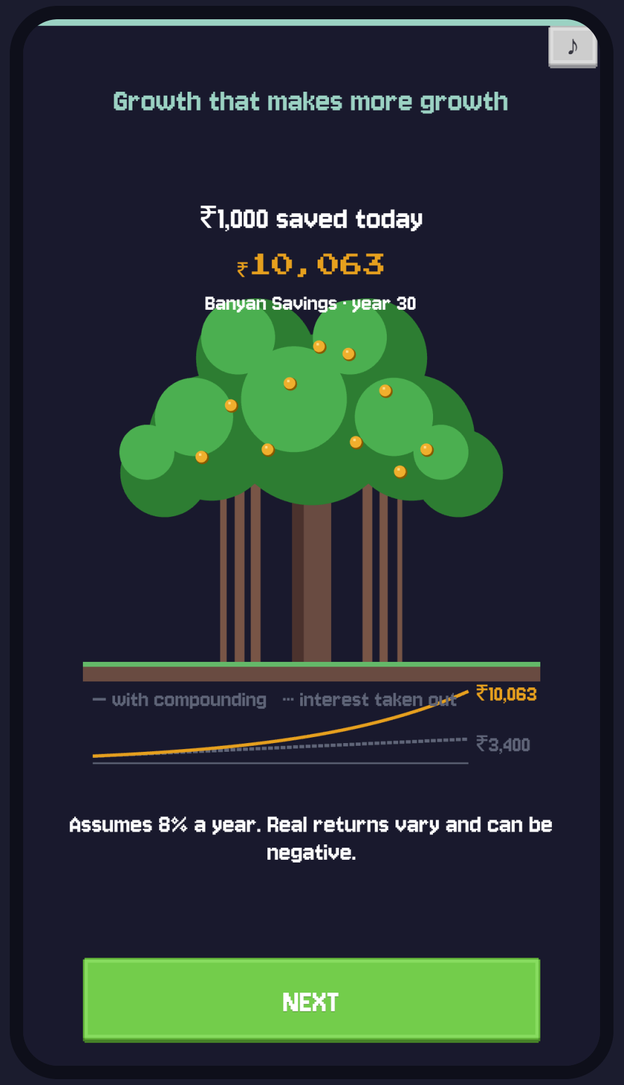
Each month opens on the whole map, by day, evening or night, with a boat and an auto on the move. Every landmark is a hub of missions.

### The Professor speaks

422 recorded voice-over lines for lessons, praise, hints and explainers, with switches for music, sound and voice.






ANALOGIES FROM HOME

# Compounding is a banyan, not a snowball

Hyderabad has no snow. The Pillalamarri banyan near Mahabubnagar is about 700 years old: its branches drop roots that become new trunks. Growth that makes more growth.

**₹10,063**
with compounding

**₹3,400**
interest taken out each year

₹1,000 saved for 30 years. Diversification is the Charminar vendor who sells umbrellas *and* water bottles, so something sells in every season.

Assumes 8% a year. Real returns vary and can be negative.

AN ECONOMY SIZED TO EACH STUDENT

# A ₹200 bat should hurt the same in the game

**in-game price = real price × ₹5,000 ÷ the student's monthly pocket money**

**Pocket money ₹5,000 a month**

## ₹200

4% of the month, in life and in the game

**Pocket money ₹2,500 a month**

## ₹400

8% of the month, in life and in the game

**Pocket money ₹1,000 a month**

## ₹1,000

20% of the month, in life and in the game

Pocket money is asked indirectly, as a weekly spending range, never an exact amount. In Stages 2 and 3 salaries take over, and rent, food and travel take 55% and 60% of income every month.

CONSEQUENCES THAT FOLLOW YOU

# Your choices come back, and so does the lesson

### Delayed effects

45 effects fire months later: the "no-cost" EMI every month, the friend who never repays, the emergency that finds no buffer.

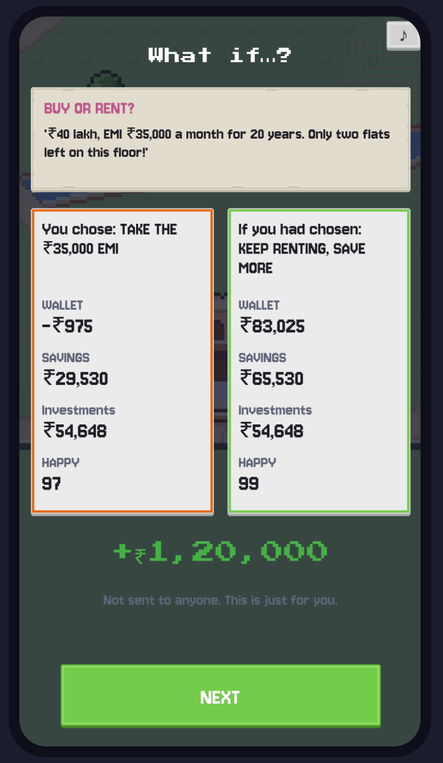
### A mistake bank

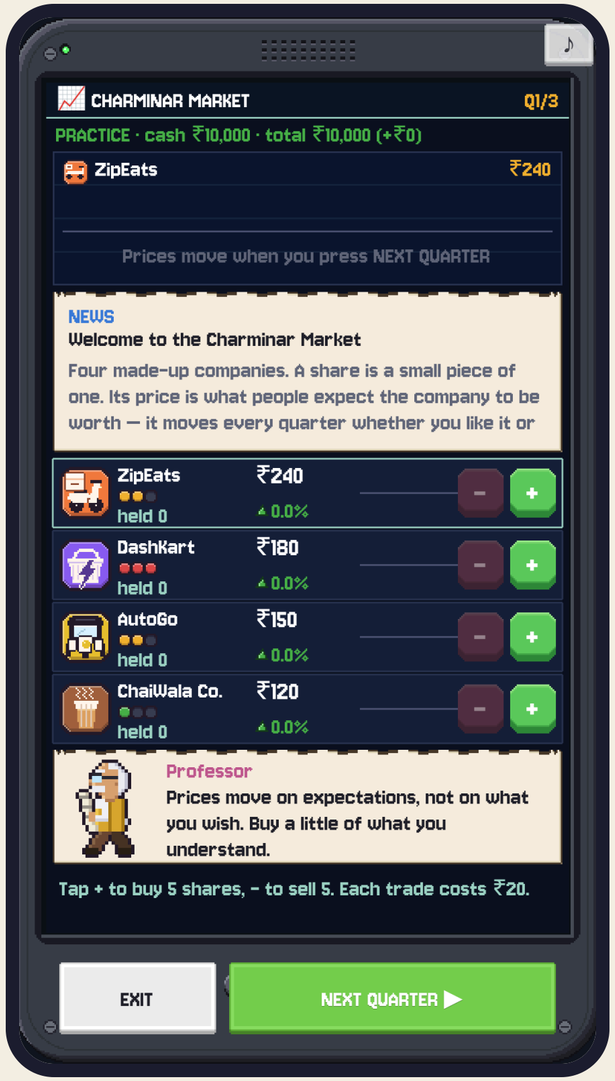
Every weak choice is banked and returns in a new disguise. It clears only when the student gets it right.

### What if?

At 60, the game replays your biggest mistake with the other choice, on the same random seed, side by side. It stays on the phone.






CHARMINAR MARKET

# A stock market that scores behaviour, not luck

Four made-up companies anyone recognises: ZipEats, DashKart, AutoGo and ChaiWala Co. Two years in eight quarters, with in-game savings only, after a practice round.

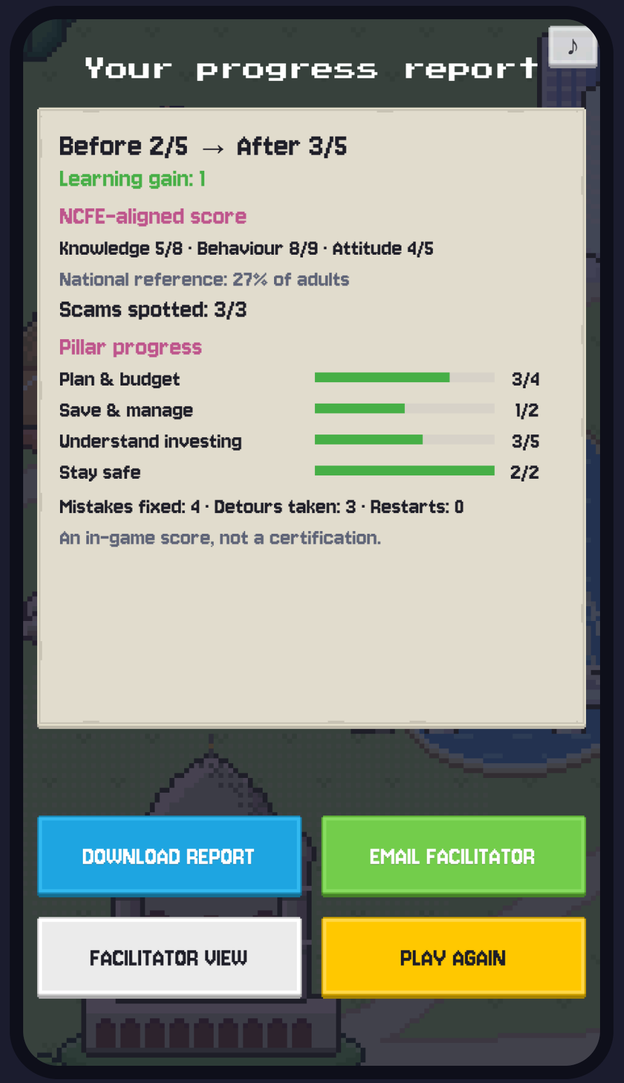
### Every run has three beats

- Monsoon: AutoGo dips
- Hot tip: a forward pumps DashKart
- Crash: everything falls, then recovers

### What gets scored

- Diversified across companies
- Held through the crash
- Refused the hot tip
- Low churn: ₹20 per trade

**Money made is shown, never scored. No real companies, no tickers, nothing recommended.**




NGO-INTEGRATED AND PRIVATE

# Built for an NGO session

1. **The facilitator sets up**
   Batch code, report email, guardian email and consent, then hands the phone over.
2. **The student plays**
   A nickname only. An offline web app that saves and resumes on shared phones.
3. **The report goes out**
   A PDF with a reason for every number, emailed to the facilitator, guardian in CC.
4. **The NGO sees progress**
   Dashboard per batch (mock today): before and after, % literate, common mistakes.

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**Never asks for an OTP, PIN, Aadhaar, or a student's phone or email. The sponsor sees aggregates only.**

RESULTS

# Tested like software, balanced like a game

**151**
unit tests passing: engine, content, mini-games, client

**9 of 9**
balance targets met by scripted players on real content

**17 of 17**
NCFE items evidenced by at least one choice

**1.2 s**
to open again on a slow network, once loaded

| Scripted player | Wallet at end | Happiness | Detours | Restarts | NCFE-literate |
|---|---|---|---|---|---|
| **Always good** | ₹0.84 L to ₹1.02 L | 98 to 99 | 0 | 0 | 100% |
| **Random** | ₹19 k to ₹44 k | 32 to 60 | 2.7 to 3.6 | 0.9 to 1.4 | 5% |
| **Always worst** | in debt, up to ₹1.77 L | 9 to 41 | 5.9 to 7.4 | 2.9 to 5.1 | 0% |

20 seeds per difficulty; every player finishes all ten months. Fast 3G with a 4× slower CPU: first visit 21 s (1.1 MB of small files, our next fix), repeat 1.2 s, offline 1.9 s.

PLAY TESTING

# Two play tests, 36 issues, a better game

Unguided, on a phone and in phone mode, at 02:15 and 06:10. Every issue got an owner; 25 of 36 were closed by 08:10, all 12 from the second test. What changed:

### Tighter economy

Careless play no longer ends rich: rent, food and travel every month.

### One teacher

The Professor praises good choices and explains only after mistakes.

### Clearer screens

A number beside the hearts, and a heading on every quiz question.

### Fairer mini-games

Longer timers (60 s and 45 s), and only items nobody can argue about.

### A fairer after-quiz

The same five concepts, asked in new situations, so gain is not memory.

### A report that explains

Charts in the game's palette and a one-line reason for every number.

### A facilitator path

Batch code, consent and a hand-over screen for NGO sessions.

### A safe market

The requested stock simulator, rebuilt to respect the guardrail.

LIMITATIONS AND NEXT STEPS

# Next: a pilot with one NGO batch

### Measure real learning

We have measured the game, not yet the students. Run the before and after quiz with a real batch, with the dashboard live.

### Ground the numbers

Prices and situations are team estimates. A short survey of 10 students on pocket money and everyday spending.

### More languages, more voice

Native review and recorded voice-over for Telugu and Hindi, Bhashini for more languages, and a faster first load.

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## Make choices. See consequences. Try again.

Team 97